

# Nonparametric Models

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## Empirical CDF

**Definition 0.1** (Empirical CDF). For observed values  $x_1, \dots, x_n$ , the **empirical cumulative distribution function (empirical CDF)** is the function

$$\hat{F}(t) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{j=1}^n I(x_j \leq t), \quad t \in \mathbb{R},$$

where  $I(A)$  is the indicator function:  $I(A) = 1$  if  $A$  holds, and  $I(A) = 0$  otherwise.

**Example 0.1** (Numerical example: empirical CDF). Suppose the observed values are 4, 1, 7, 3, so  $n = 4$ . Two of the four values (1 and 3) are at most 3.5, so:

$$\begin{aligned} \hat{F}(3.5) &= \frac{1}{4}(I(4 \leq 3.5) + I(1 \leq 3.5) + I(7 \leq 3.5) + I(3 \leq 3.5)) && \text{(definition of } \hat{F}) \\ &= \frac{1}{4}(0 + 1 + 0 + 1) && \text{(evaluate each indicator)} \\ &= 0.5 && \text{(arithmetic)} \end{aligned}$$

Similarly,  $\hat{F}(0) = 0$ , because no value is at most 0, and  $\hat{F}(7) = 1$ , because every value is at most 7.

## Order statistics

**Definition 0.2** (Order statistics). For observed values  $x_1, \dots, x_n$ , the **order statistics** are the same values sorted in increasing order, written

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

So  $x_{(1)}$  is the smallest value and  $x_{(n)}$  is the largest.

**Example 0.2** (Numerical example: order statistics). If the observed values are 4, 1, 7, 3, then the order statistics are  $x_{(1)} = 1$ ,  $x_{(2)} = 3$ ,  $x_{(3)} = 4$ , and  $x_{(4)} = 7$ .

## Sample quantiles

**Definition 0.3** (Sample quantile function). Given an **empirical CDF**  $\hat{F}$ , the **sample quantile function** (also called the **empirical quantile function**) is

$$\hat{Q}(p) \stackrel{\text{def}}{=} \inf\{t : \hat{F}(t) \geq p\}, \quad 0 < p \leq 1.$$

For a given  $p$ , the value  $\hat{Q}(p)$  is the **sample  $p$  quantile**, also called the **sample 100 $p$ th percentile**.

The **CDF**  $F$  and the **quantile function**  $Q$  describe a probability distribution, while  $\hat{F}$  and  $\hat{Q}$  describe a sample, and serve as estimates of  $F$  and  $Q$ .

**Theorem 0.1** (Order-statistics form of the sample quantile). *If  $x_{(1)} \leq \cdots \leq x_{(n)}$  are the order statistics and  $i \in \{1, \dots, n\}$ , then  $\hat{Q}(p) = x_{(i)}$  for every  $p$  with  $(i-1)/n < p \leq i/n$ . In particular,  $\hat{Q}(i/n) = x_{(i)}$ .*

### **i** Proof

*Proof.* Fix  $p$  with  $(i-1)/n < p \leq i/n$ .

For any  $t < x_{(i)}$ , the  $n-i+1$  values  $x_{(i)}, \dots, x_{(n)}$  all exceed  $t$ , so at most  $i-1$  values are at most  $t$ , and:

$$\begin{aligned} \hat{F}(t) &\leq \frac{i-1}{n} && \text{(at most } i-1 \text{ indicators equal 1)} \\ &< p && \text{(choice of } p) \end{aligned}$$

So no  $t < x_{(i)}$  belongs to the set  $\{t : \hat{F}(t) \geq p\}$ .

At  $t = x_{(i)}$ , the  $i$  values  $x_{(1)}, \dots, x_{(i)}$  are all at most  $x_{(i)}$ , so:

$$\begin{aligned}\hat{F}(x_{(i)}) &\geq \frac{i}{n} \quad (\text{at least } i \text{ indicators equal } 1) \\ &\geq p \quad (\text{choice of } p)\end{aligned}$$

So  $x_{(i)}$  belongs to the set, and it is the smallest member of the set, because no smaller  $t$  belongs to it. Therefore  $\hat{Q}(p) = x_{(i)}$ .  $\square$

**Example 0.3** (Numerical example: sample quantiles). Using the same data 4, 1, 7, 3, with order statistics 1, 3, 4, 7, Theorem 0.1 says the sample quantile function is a step function:

- $\hat{Q}(p) = 1$  for  $0 < p \leq 1/4$ ;
- $\hat{Q}(p) = 3$  for  $1/4 < p \leq 1/2$ ;
- $\hat{Q}(p) = 4$  for  $1/2 < p \leq 3/4$ ;
- $\hat{Q}(p) = 7$  for  $3/4 < p \leq 1$ .

So  $\hat{Q}(0.5) = 3$  and  $\hat{Q}(0.9) = 7$ . R's `quantile()` function computes this definition when called with `type = 1`:

```
quantile(c(4, 1, 7, 3), probs = c(0.5, 0.9), type = 1)
#> 50% 90%
#>   3   7
```

Other sources, and other software defaults, define sample quantiles differently, mostly by interpolating between order statistics. R's `quantile()` offers nine definitions through its `type` argument, and its default (`type = 7`) interpolates, so `quantile(c(4, 1, 7, 3), 0.5)` returns 3.5, not 3. The usual sample median is another interpolated quantile: for an even number of observations, it averages the two middle order statistics. These notes use Definition 0.3, which always returns one of the observed values.

## The empirical CDF and quantile function as generalized inverses

Both functions in Figure 1 are step functions, so neither is one-to-one. Each flat piece of one function corresponds to a jump of the other: for example,  $\hat{F}(t) = 0.5$  for every  $t \in [3, 4)$ , and  $\hat{Q}$  jumps from 3 to 4 at  $p = 0.5$ . So  $\hat{Q}$  is a *generalized* inverse of  $\hat{F}$ , not an inverse in the usual sense:  $\hat{Q}(\hat{F}(t)) \leq t$  for every  $t \geq x_{(1)}$ , with equality only at the observed values.

```

x <- c(4, 1, 7, 3)
n <- length(x)
x_ord <- sort(x)
p_ord <- seq_len(n) / n
ecdf_x_padding <- 0.8
x_min <- min(x_ord) - ecdf_x_padding
x_max <- max(x_ord) + ecdf_x_padding

x_left <- c(x_min, x_ord)
x_right <- c(x_ord, x_max)
f_levels <- c(0, p_ord)

plot(
  0,
  0,
  type = "n",
  xlim = c(x_min, x_max),
  ylim = c(0, 1.05),
  xlab = "t",
  ylab = expression(hat(plain("F"))(t))
)
segments(x_left, f_levels, x_right, f_levels, lwd = 2, col = "blue")
points(x_ord, f_levels[seq_len(n)], pch = 1, col = "blue")
points(x_ord, f_levels[seq_len(n) + 1], pch = 19, col = "blue")

```



(a) Empirical CDF, horizontal pieces only. Closed circles mark included endpoints; open circles mark excluded endpoints.

```

eqf_y_padding <- 0.5
q_left <- c(0, p_ord[-n])
q_right <- p_ord
plot(
  0,
  0,
  type = "n",
  xlim = c(0, 1),
  ylim = c(min(x_ord) - eqf_y_padding, max(x_ord) + eqf_y_padding),
  xlab = "p",
  ylab = expression(hat(Q)(p))
)
segments(q_left, x_ord, q_right, x_ord, lwd = 2, col = "blue")
points(q_left, x_ord, pch = 1, col = "blue")
points(q_right, x_ord, pch = 19, col = "blue")

```



(b) Sample quantile function, horizontal pieces only. Open circles mark excluded left endpoints; closed circles mark included right endpoints.

Figure 1: The empirical CDF and the sample quantile function for the data 4, 1, 7, 3 (order statistics 1, 3, 4, 7).